

Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

ANSWER KEY



Compound angle formula for cosine

- 1 Use the compound angle formula $\cos(A + B) = \cos A \cos B - \sin A \sin B$ to find the exact value of $\cos 75^\circ$, writing $75^\circ = 45^\circ + 30^\circ$.

$\cos 75^\circ = \cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ$. Since $\cos 45^\circ = \sin 45^\circ = \sqrt{2}/2$, $\cos 30^\circ = \sqrt{3}/2$, and $\sin 30^\circ = 1/2$, this gives $\cos 75^\circ = (\sqrt{6} - \sqrt{2})/4$.

Compound angle formula for sine

- 2 Use the compound angle formula $\sin(A - B) = \sin A \cos B - \cos A \sin B$ to find the exact value of $\sin 15^\circ$, writing $15^\circ = 45^\circ - 30^\circ$.

$\sin 15^\circ = \sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ$. Since $\sin 45^\circ = \cos 45^\circ = \sqrt{2}/2$, $\cos 30^\circ = \sqrt{3}/2$, and $\sin 30^\circ = 1/2$, this gives $\sin 15^\circ = (\sqrt{6} - \sqrt{2})/4$. (This matches $\cos 75^\circ$ above, as expected since $\sin 15^\circ = \cos 75^\circ$.)

Compound angle formula for tangent

- 3 The compound angle formula for tangent is $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$. Use it with $A = 45^\circ$ and $B = 30^\circ$ to find the exact value of $\tan 15^\circ$, giving your answer in simplified surd form.

Substituting $\tan 45^\circ = 1$ and $\tan 30^\circ = 1/\sqrt{3}$: $\tan 15^\circ = (1 - 1/\sqrt{3})/(1 + 1/\sqrt{3})$. Multiplying top and bottom by $\sqrt{3}$: $\tan 15^\circ = (\sqrt{3} - 1)/(\sqrt{3} + 1)$. Rationalising by multiplying top and bottom by $(\sqrt{3} - 1)$: the numerator becomes $(\sqrt{3} - 1)^2 = 4 - 2\sqrt{3}$ and the denominator becomes $(\sqrt{3} + 1)(\sqrt{3} - 1) = 2$. So $\tan 15^\circ = (4 - 2\sqrt{3})/2 = 2 - \sqrt{3}$.

Proving identities using the compound angle formulas

- 4 Prove that $\sin(A + B) + \sin(A - B) = 2\sin A \cos B$.

Expand both: $\sin(A + B) = \sin A \cos B + \cos A \sin B$ and $\sin(A - B) = \sin A \cos B - \cos A \sin B$. Adding them, the $\cos A \sin B$ terms cancel: $\sin(A + B) + \sin(A - B) = 2\sin A \cos B$, as required.

- 5 Prove that $\cos(A + B)\cos(A - B) = \cos^2 A - \sin^2 B$.

Expand: $\cos(A + B) = \cos A \cos B - \sin A \sin B$ and $\cos(A - B) = \cos A \cos B + \sin A \sin B$. Their product is a difference of squares: $\cos^2 A \cos^2 B - \sin^2 A \sin^2 B$. Using $\cos^2 B = 1 - \sin^2 B$: this equals $\cos^2 A(1 - \sin^2 B) - \sin^2 A \sin^2 B = \cos^2 A - \sin^2 B(\cos^2 A + \sin^2 A) = \cos^2 A - \sin^2 B$, as required.

Solving a trigonometric equation using a compound angle formula

- 6 Solve $\sin(x + 30^\circ) = \cos x$ for $0^\circ \leq x \leq 360^\circ$.

Expand the left side first: $\sin(x + 30^\circ) = \sin x \cos 30^\circ + \cos x \sin 30^\circ$, which is

$(\sqrt{3}/2)\sin x + (1/2)\cos x$. Setting this equal to $\cos x$ and multiplying every term by 2:

$\sqrt{3}\sin x + \cos x = 2\cos x$. Rearranging: $\sqrt{3}\sin x = \cos x$, so $\tan x = 1/\sqrt{3}$. Since this has period 180° :
 $x = 30^\circ$ or $x = 210^\circ$.

Evaluating reciprocal trigonometric functions from a given ratio

- 7 Given $\cos \theta = \frac{3}{5}$ with θ in Quadrant I, find the exact values of $\sec \theta$, $\csc \theta$, and $\cot \theta$.

In Quadrant I all ratios are positive. By the Pythagorean identity, $\sin \theta = \sqrt{1 - 9/25} = 4/5$. Then
 $\sec \theta = 1/\cos \theta = 5/3$, $\csc \theta = 1/\sin \theta = 5/4$, and $\cot \theta = \cos \theta / \sin \theta = 3/4$.

Reciprocal trigonometric functions in another quadrant

- 8 Given $\tan \theta = -2$ with θ in Quadrant II, find the exact values of $\sec \theta$, $\csc \theta$, and $\cot \theta$.

In Quadrant II, $\sin \theta > 0$ and $\cos \theta < 0$. Treating $\tan \theta = -2$ as opposite = 2, adjacent = -1, the reference triangle has hypotenuse $\sqrt{5}$. So $\sin \theta = 2/\sqrt{5}$ and $\cos \theta = -1/\sqrt{5}$. Then
 $\sec \theta = 1/\cos \theta = -\sqrt{5}$, $\csc \theta = 1/\sin \theta = \sqrt{5}/2$, and $\cot \theta = 1/\tan \theta = -1/2$.

Deriving $1 + \tan^2 \theta = \sec^2 \theta$

- 9 Starting from the Pythagorean identity $\sin^2 \theta + \cos^2 \theta = 1$, divide through by $\cos^2 \theta$ to derive $1 + \tan^2 \theta = \sec^2 \theta$.

Dividing every term by $\cos^2 \theta$: $\frac{\sin^2 \theta}{\cos^2 \theta} + \frac{\cos^2 \theta}{\cos^2 \theta} = \frac{1}{\cos^2 \theta}$, which is exactly $\tan^2 \theta + 1 = \sec^2 \theta$.

Deriving $1 + \cot^2 \theta = \csc^2 \theta$

- 10 Starting from the Pythagorean identity $\sin^2 \theta + \cos^2 \theta = 1$, divide through by $\sin^2 \theta$ to derive $1 + \cot^2 \theta = \csc^2 \theta$.

Dividing every term by $\sin^2 \theta$: $\frac{\sin^2 \theta}{\sin^2 \theta} + \frac{\cos^2 \theta}{\sin^2 \theta} = \frac{1}{\sin^2 \theta}$, which is exactly $1 + \cot^2 \theta = \csc^2 \theta$.

Simplifying an expression using reciprocal identities

- 11 Simplify $\sec \theta - \cos \theta$, expressing your answer in terms of $\sin \theta$ and $\tan \theta$.

$$\sec \theta - \cos \theta = \frac{1}{\cos \theta} - \cos \theta = \frac{1 - \cos^2 \theta}{\cos \theta} = \frac{\sin^2 \theta}{\cos \theta} = \sin \theta \cdot \frac{\sin \theta}{\cos \theta} = \sin \theta \tan \theta.$$

Mixed application: combining reciprocal identities across quadrants

- 12** Given $\sec \theta = 13/5$ with θ in Quadrant IV, find the exact value of $\sin \theta + \cot \theta$.
 $\cos \theta = 5/13$. By the Pythagorean identity, $\sin^2 \theta = 1 - 25/169 = 144/169$. Since θ is in Quadrant IV, $\sin \theta = -12/13$. Then $\cot \theta = \cos \theta / \sin \theta = (5/13) / (-12/13) = -5/12$. Adding a common denominator of 156: $\sin \theta + \cot \theta = -144/156 - 65/156 = -209/156$.
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Compound angle formula for sine (sum form)

- 13** Use the compound angle formula $\sin(A + B) = \sin A \cos B + \cos A \sin B$ to find the exact value of $\sin 75^\circ$, writing $75^\circ = 45^\circ + 30^\circ$.
 $\sin 75^\circ = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ$. Since $\sin 45^\circ = \cos 45^\circ = \sqrt{2}/2$, $\cos 30^\circ = \sqrt{3}/2$, and $\sin 30^\circ = 1/2$, this gives $\sin 75^\circ = (\sqrt{6} + \sqrt{2})/4$.
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Compound angle formula for tangent (sum form)

- 14** Use $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$ with $A = 45^\circ$ and $B = 30^\circ$ to find the exact value of $\tan 75^\circ$, giving your answer in simplified surd form.
Substituting $\tan 45^\circ = 1$ and $\tan 30^\circ = 1/\sqrt{3}$: $\tan 75^\circ = (1 + 1/\sqrt{3}) / (1 - 1/\sqrt{3})$. Multiplying top and bottom by $\sqrt{3}$: $\tan 75^\circ = (\sqrt{3} + 1) / (\sqrt{3} - 1)$. Rationalising by multiplying top and bottom by $(\sqrt{3} + 1)$: the numerator becomes $(\sqrt{3} + 1)^2 = 4 + 2\sqrt{3}$ and the denominator becomes $(\sqrt{3} - 1)(\sqrt{3} + 1) = 2$. So $\tan 75^\circ = (4 + 2\sqrt{3})/2 = 2 + \sqrt{3}$.
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Finding an exact value in Quadrant II

- 15** Use the compound angle formula $\cos(A + B) = \cos A \cos B - \sin A \sin B$ to find the exact value of $\cos 105^\circ$, writing $105^\circ = 60^\circ + 45^\circ$.
 $\cos 105^\circ = \cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ$. Since $\cos 60^\circ = 1/2$, $\cos 45^\circ = \sin 45^\circ = \sqrt{2}/2$, and $\sin 60^\circ = \sqrt{3}/2$, this gives $\cos 105^\circ = (\sqrt{2} - \sqrt{6})/4$ — negative, as expected since 105° is in Quadrant II.
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Deriving the double-angle formula from the compound angle formula

- 16** By setting $A = B = \theta$ in $\sin(A + B) = \sin A \cos B + \cos A \sin B$, derive the double-angle formula $\sin 2\theta = 2\sin \theta \cos \theta$.
Setting $A = B = \theta$: $\sin(\theta + \theta) = \sin \theta \cos \theta + \cos \theta \sin \theta$. The right side has two identical terms, so $\sin 2\theta = 2\sin \theta \cos \theta$, as required.
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A second identity from the compound angle formulas

- 17 Prove that $\cos(A - B) - \cos(A + B) = 2\sin A \sin B$.

Expand both: $\cos(A - B) = \cos A \cos B + \sin A \sin B$ and $\cos(A + B) = \cos A \cos B - \sin A \sin B$.

Subtracting, the $\cos A \cos B$ terms cancel: $\cos(A - B) - \cos(A + B) = 2\sin A \sin B$, as required.

Solving another trigonometric equation using a compound angle formula

- 18 Solve $\sin(x + 60^\circ) = \cos x$ for $0^\circ \leq x \leq 360^\circ$.

Expand the left side: $\sin(x + 60^\circ) = \sin x \cos 60^\circ + \cos x \sin 60^\circ$, which is $(1/2)\sin x + (\sqrt{3}/2)\cos x$.

Setting this equal to $\cos x$ and multiplying every term by 2: $\sin x + \sqrt{3}\cos x = 2\cos x$. Rearranging:

$\sin x = (2 - \sqrt{3})\cos x$, so $\tan x = 2 - \sqrt{3}$, which (as found earlier in this worksheet) equals $\tan 15^\circ$.

Since this has period 180° : $x = 15^\circ$ or $x = 195^\circ$.

A quick simplification from the Pythagorean identity

- 19 Simplify $(\sec \theta + \tan \theta)(\sec \theta - \tan \theta)$.

This is a difference of squares: $\sec^2 \theta - \tan^2 \theta$. By the identity $1 + \tan^2 \theta = \sec^2 \theta$, this equals 1.

Reciprocal trigonometric functions in Quadrant III

- 20 Given $\sec \theta = -13/12$ with θ in Quadrant III, find the exact values of $\sin \theta$, $\cos \theta$, and $\tan \theta$.

$\cos \theta = 1/\sec \theta = -12/13$. In Quadrant III, $\sin \theta$ is also negative. By the Pythagorean identity,

$\sin^2 \theta = 1 - 144/169 = 25/169$, so $\sin \theta = -5/13$. Then

$\tan \theta = \sin \theta / \cos \theta = (-5/13)/(-12/13) = 5/12$.

Connecting a co-function identity to the reciprocal functions

- 21 Using the compound angle formula for cosine, show that $\cos(90^\circ - \theta) = \sin \theta$, and hence prove that $\sec(90^\circ - \theta) = \csc \theta$.

$\cos(90^\circ - \theta) = \cos 90^\circ \cos \theta + \sin 90^\circ \sin \theta = 0 + \sin \theta = \sin \theta$. Then

$\sec(90^\circ - \theta) = 1/\cos(90^\circ - \theta) = 1/\sin \theta = \csc \theta$, as required.

Solving an equation involving a reciprocal function

- 22 Solve $\csc \theta = 2$ for $0^\circ \leq \theta \leq 360^\circ$.

$\csc \theta = 2 \Rightarrow \sin \theta = 1/2$. Since sine is positive in Quadrants I and II: $\theta = 30^\circ$ or $\theta = 150^\circ$.

Proving a reciprocal identity from scratch

23 Prove that $\tan \theta + \cot \theta = \sec \theta \csc \theta$.

$\tan \theta + \cot \theta = \sin \theta / \cos \theta + \cos \theta / \sin \theta$. Combining over a common denominator, this is $(\sin^2 \theta + \cos^2 \theta) / (\sin \theta \cos \theta)$. Since $\sin^2 \theta + \cos^2 \theta = 1$, this equals $1 / (\sin \theta \cos \theta)$. On the other side, $\sec \theta \csc \theta = (1 / \cos \theta)(1 / \sin \theta) = 1 / (\sin \theta \cos \theta)$. The two sides match, as required.

A capstone exact-value application

24 Find the exact value of $\cos 15^\circ$ (writing $15^\circ = 45^\circ - 30^\circ$), then use it together with $\cos 75^\circ = (\sqrt{6} - \sqrt{2})/4$ found earlier to compute $\cos 75^\circ - \cos 15^\circ$ exactly.

$\cos 15^\circ = \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ$. This equals $(\sqrt{2} + \sqrt{6})/4$. Now subtract:
 $\cos 75^\circ - \cos 15^\circ = (\sqrt{6} - \sqrt{2})/4 - (\sqrt{6} + \sqrt{2})/4$. The $\sqrt{6}$ terms cancel, leaving $-2\sqrt{2}/4$. This simplifies to $-\sqrt{2}/2$.